Spark App

$x = 10$ → Step 1 (Read)

$x = x + 20$ → Step2 (Transform)

$x = 30$ → Step3 (Write in Database)

10GB

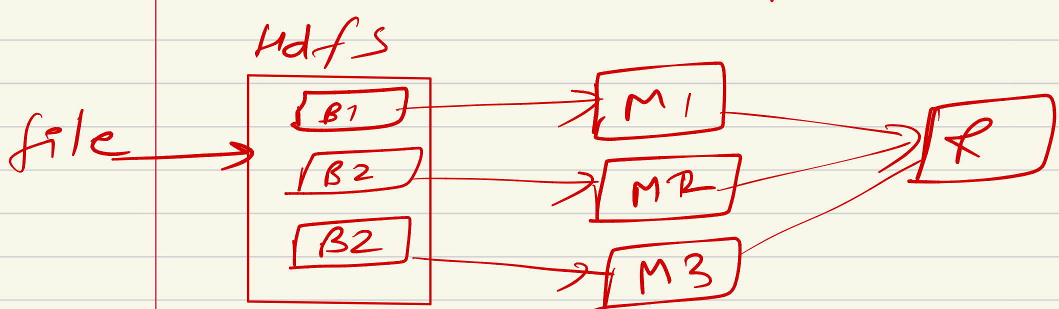
emp_data = spark.read_csv("emp.csv")

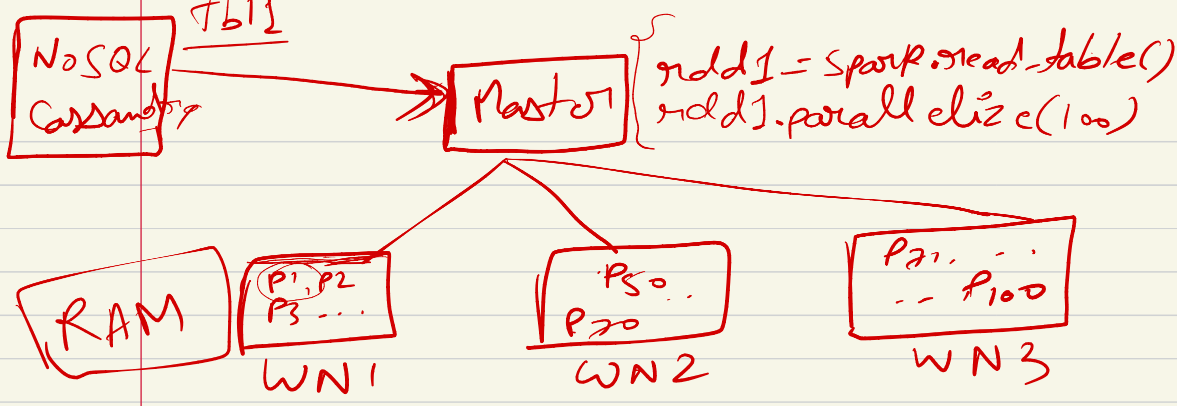
RDD →

Partition

Tokopedia

How do you make data partitioned in Spark?





1TB

1TB

1TP

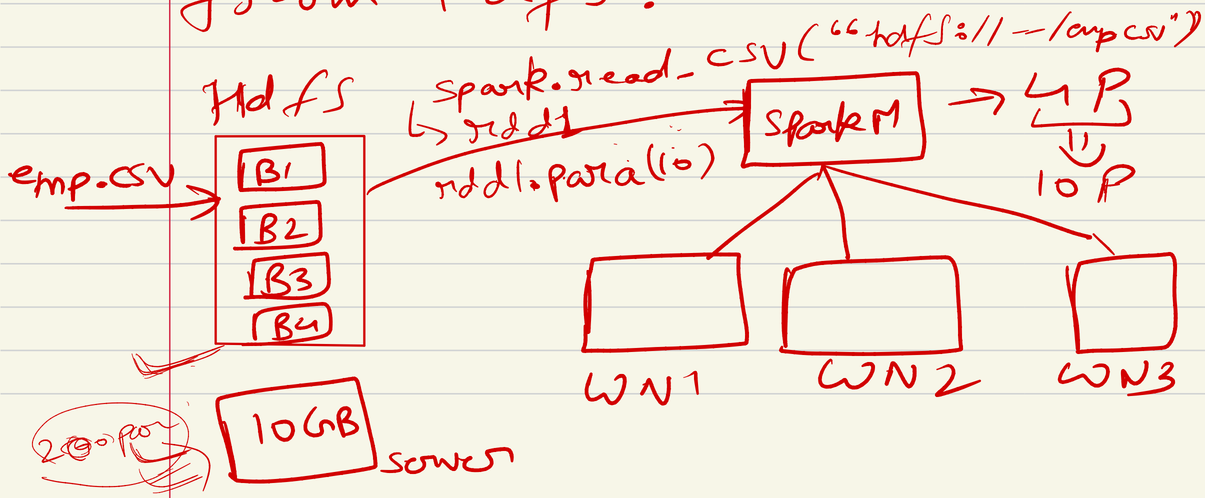
Block = Partition

(Hadoop) (Spark)

replication

??

Ex: Spark app to process data from Hdfs?



Partition

→ Parallelize (n)

→ Read data from any such source where data is already distributed

"emp.csv"

emp_N	emp_id	salary

n = 10

rdd1 = spark.read_csv("emp.csv")

rdd2 = rdd1.withColumn("new_sal", 20%)

n = n + 20

rdd3 = rdd2.func()

rdd1

↓

rdd2

DAG

⇓

Lineage Graph

rdd3

Transformations

Actions

4 Byte

a

public int a ;

Declaration Definition

Public int a=10

step1 (Read)

step2 (Group By)

step3 (Join)

Counting, selecting, write

first( ) Action

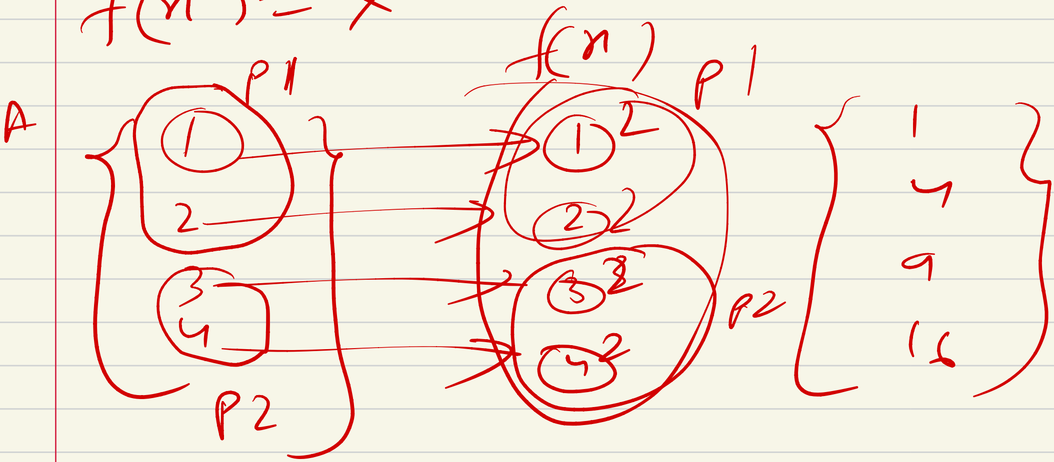
RDD

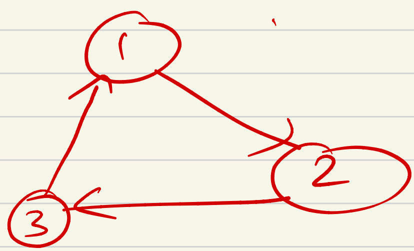
↓

Dataframe ⇒ Structured RDD

$A = \{1, 2, 3, 4\}$

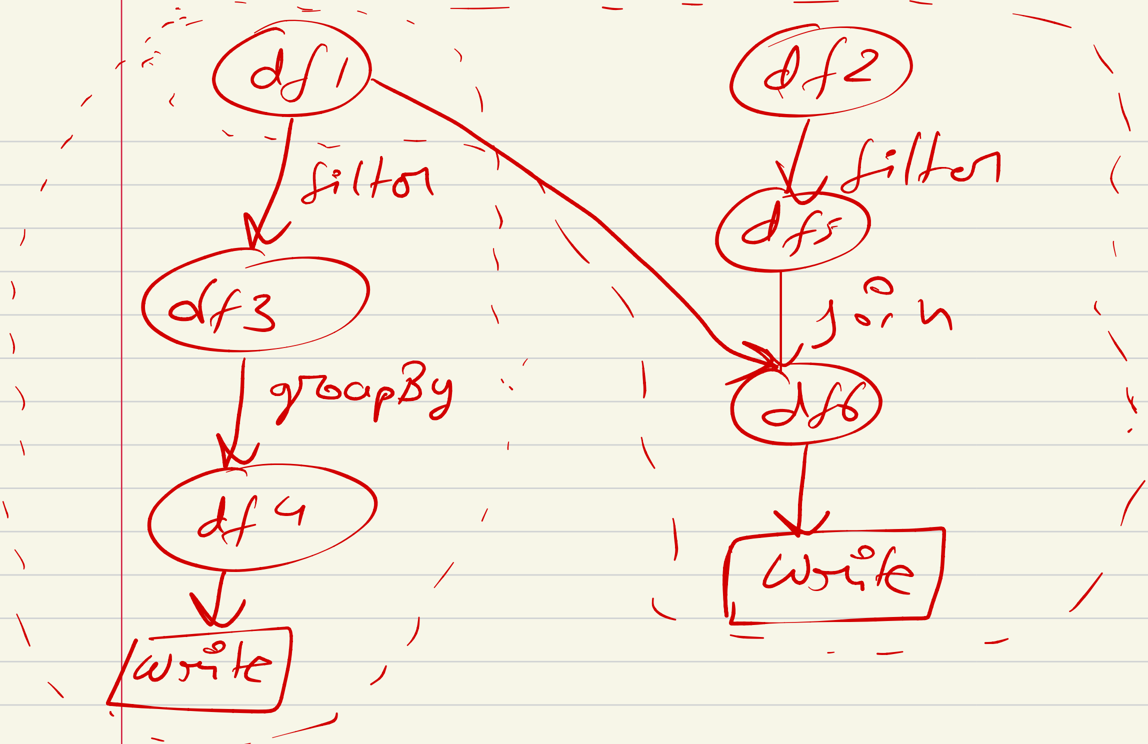
$f(x) = x^2$





$$\text{rdd1} \xrightarrow{\text{groupBy}} \text{rdd2}$$

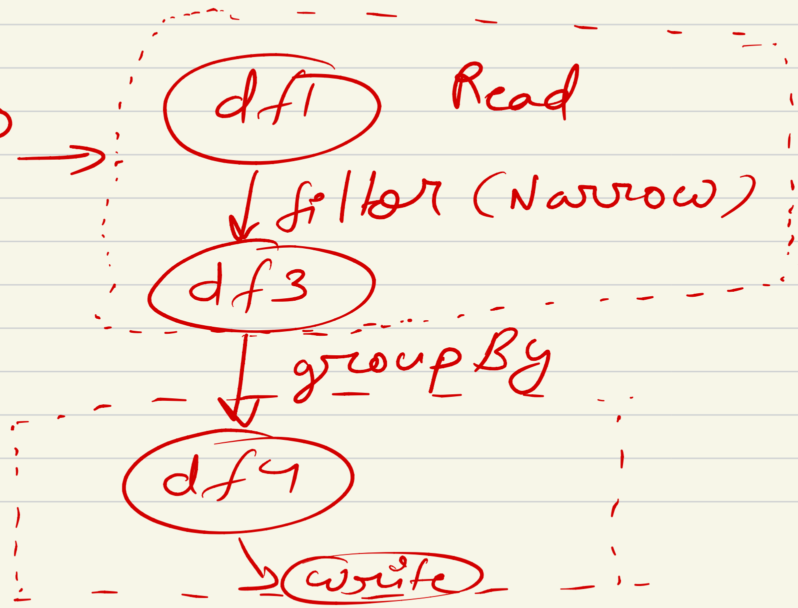
```
df1 = read_csv("emp_csv")
df2 = read_csv("dept_csv")

df3 = df1.filter("INDIA")
df4 = df3.groupBy("Dept_wise")

df4.write(hdfs)

df5 = df2.filter("IT")
df6 = df1.join(df5)
df6.write(hdfs)
```



DAG 0
JOB 0

DAG 1
JOB 1

stage 0



JOB 0

stage 1

emp, dept, region